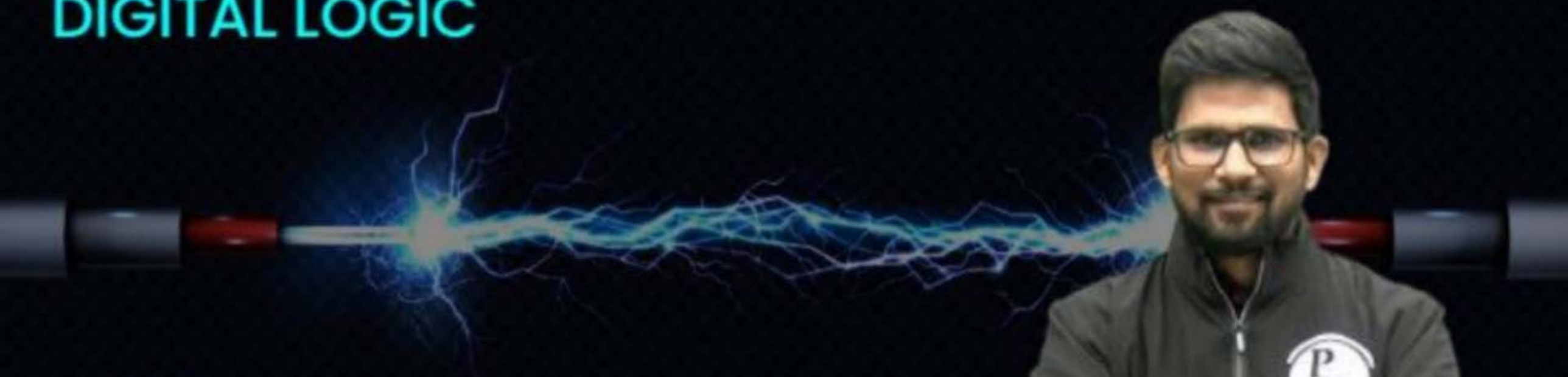
COMPUTER SCIENCE & IT

DIGITAL LOGIC



Lecture No. 03

BOOLEAN THEOREMS AND GATES



By- Chandan Gupta Sir



Theorems



Theorems & imp. basics.

H.W

Q.1 $A\bar{B} + A\bar{C} + BC$ simplifies to

a. $(A+B)(A+C)$

b. $(A+\bar{B})(A+\bar{C})$

c. $(\bar{A}+B)(A+\bar{C})$

d. $(A+B)(B+C)$

Ans: option (A)

$[A + B\bar{C}\cdot C]$

$[A+B+\bar{C}]$

$= A[A+B+\bar{C}] = A$

$= A + AB + AC$

$= A[1 + B + C] = A \cdot 1 = A$

Q2 $A + \bar{B} + \bar{C} + AD + \bar{A}B$ simplifies to

a. 1

b. $A+B+C+D$

c. 0

d. $(A+\bar{B}+\bar{C}+D)$

Ans: option (a)

$(A+\bar{B}) + \bar{C} + AD + \bar{A}B$

$\bar{x} + \bar{C} + AD + x$

$= 1$

Q3 $(A+B)(B+\bar{C})(A+B+C)$ simplifies to

Ans ⟶ option (B)

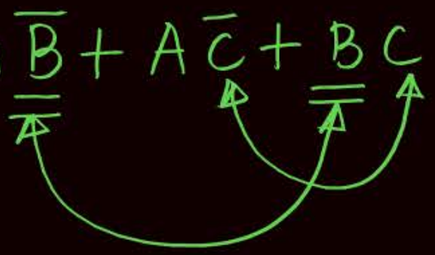
a. $A+B\bar{C}$ b. $B+A\bar{C}$ c. $C+A\bar{B}$ d. $AB+C$

Q.4 $(A+B\bar{C})(A+C)(A+B+\bar{C})$ Ans ⟶ option (a)

equal to

a. A b. $(A+B)(A+\bar{C})$ c. $A+BC$ d. None of this

- $A\bar{B} + A\bar{C} + BC$



$= A[\bar{B} + \bar{C}] + BC$

$BC = x \rightarrow A\bar{x} + x$

$\overline{BC} = \bar{x}$

$\bar{B} + \bar{C} = \bar{x}$

$= x + (A \cdot \bar{x})$

$= (x + A) \cdot (x + \bar{x})$

$= (x + A)$

$= A + (BC)$

$= (A + B) \cdot (A + C)$

$A + \bar{B} + \bar{C} + AD + \bar{A}B$

$= A + (\bar{A} \cdot B) + \bar{B} + \bar{C} + AD$

$= (A + \bar{A}) \cdot (A + B) + \bar{B} + \bar{C} + AD$

$= (A + B) + \bar{B} + \bar{C} + AD$

$= 1$

- $(A + B + 0)(B + \bar{C})(A + B + C)$

$= [(A + B) + 0 \cdot C][B + \bar{C}]$

$= (A + B)(B + \bar{C}) = B + A \cdot \bar{C}$

MSQ

#Q. $\overline{AB} + B\overline{C}$ simplifies to [is/are]

a. $\overline{A}+B+\overline{C}$ ✗

b. $\overline{A}+\overline{B}+\overline{C}$ ✓

c. $\overline{AB}+\overline{C}$ ✓

d. $AB+BC$ ✗

(B, C)

$\overline{AB} + B\overline{C}$

$= \overline{A} + \overline{B} + B\cdot\overline{C}$

$= \overline{A} + (\overline{B}+B)\cdot(\overline{B}+\overline{C})$

$= \overline{A}+\overline{B}+\overline{C} = \overline{A} + \overline{BC}$

$= \overline{AB} + \overline{C}$

$= \overline{A\cdot B\cdot C}$

#Q. $(\overline{A}+\overline{B})C + ABCD$ is equal to

a. $(\overline{A}+\overline{B}+D)C$ ✓ b. $(\overline{A}+\overline{B}+C)D$

c. $\overline{A}B + (B+C)D$ d. $(\overline{A}+\overline{B}+\overline{C})D$

$\Rightarrow (\overline{A}+\overline{B})C + ABCD$

$\overline{P}C + PCD$ $\quad AB = P$, $\overline{AB} = \overline{P}$

$= C[\overline{P}+(P\cdot D)]$

$= C[(\overline{P}+P)\cdot(\overline{P}+D)]$

$= C(\overline{P}+D)$

$= C[\overline{AB}+D] = C[\overline{A}+\overline{B}+D]$

Question

$A + BC + \bar{A}C$ is equal to

(a) $(A + C)$

(b) $(A + B)(B + C)$

(c) $(A + B)(\bar{A} + C)$

(d) $C(\bar{A} + B)$

$A \cdot 1 + BC + \bar{A}C$

$= A + (\bar{A} \cdot C) + BC$

$= (A + \bar{A}) \cdot (A + C) + BC$

$= (A + C) + BC$

$= A + C(1 + B)$

$= (A + C)$

Question

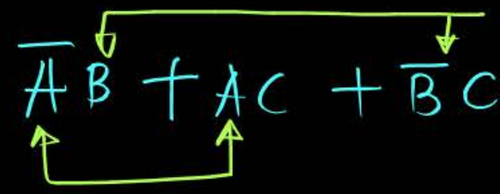
$\bar{A}B + AC + \bar{B}C$ is equivalent to

(a) $\bar{A}B + AC$

(b) $\bar{A}B + C$

(c) $AC + \bar{B}C$

(d) $\bar{A}B + \bar{B}C$



$\bar{x} + xC$

$= \bar{A}B + (A + \bar{B})C$

$= (P + \bar{P} \cdot C)$

$= (P + \bar{P}) \cdot (P + C)$

$= P + C$

$= (\bar{A} \cdot B + C) = (\bar{A} + C) \cdot (B + C)$

$\bar{A}B = P$

$\overline{\bar{A}B} = \bar{P}$

$(A + \bar{B}) = \bar{P}$

Question

$\overline{A\bar{B}(\bar{A}B + \bar{B}C + A\bar{B}D + A\bar{B}\bar{D})}$ is equal to

(a) $(\bar{A} + B)$

(b) $(\bar{A} + B)(B + \bar{C})$

(c) $(\bar{A} + B)(A + \bar{B})$

(d) $\bar{A}B + \bar{B}D$

$\Rightarrow A\bar{B}[D+\bar{D}] = A\bar{B} = P$

$\overline{P[P + \text{anything}]}$

$\overline{P} = \overline{A\bar{B}} = (\bar{A}+B)$

$P[P + \text{anything}] = P$

$= (P \cdot P + P \cdot \text{anything})$

$= P + P \cdot \text{anything}$

$= P[1 + \text{anything}]$

$= P \cdot 1 = P$

$$A\bar{B}C\left[AB + BC + AC + \bar{A}\bar{B}\bar{C} + ABD + \bar{A}CD\right]$$

$$\bar{B} \cdot AC \cdot [AC + \text{anything}]$$

$$= \bar{B} \cdot AC$$

$$= A\bar{B}C$$

Question

$\overline{(\bar{A}+\bar{B})\cdot(\bar{B}+\bar{C})}$ is equal to

(a) $\bar{B}(A+C)$

(b) $A(B+C)$

(c) $B(A+C)$

(d) $C(A+B)$

$= \overline{(\bar{B})+\bar{A}\cdot\bar{C}} = B\cdot(\overline{\bar{A}\cdot\bar{C}})$

$= B\cdot(A+C)$

$= \overline{(\bar{A}+\bar{B})} + \overline{(\bar{B}+\bar{C})}$

$= A\cdot B + B\cdot C$

$= B[A+C]$

Question

$\bar{A}\bar{B} + AC + \bar{B}C$ is equivalent to

(a) $(A + \bar{B}) \cdot (\bar{A}\bar{B} + C)$

(b) $\bar{A}\bar{B} + AC$

(c) $AC + \bar{B}C$

(d) $\bar{A}\bar{B} + \bar{B}C$

$\bar{A}\bar{B} + AC + \bar{B}C = \bar{A}\bar{B} + AC$

$\bar{B}C$ redundant

Transpose Theorem:

- $(A+B)(\bar{A}+C) = AC+\bar{A}B$

$= A\cdot\bar{A} + AC + \bar{A}B + BC$

$= AC + \bar{A}B + BC$

BC → Redundant

- $(A+B)(\bar{A}+\bar{B}) = A\bar{B}+\bar{A}B$

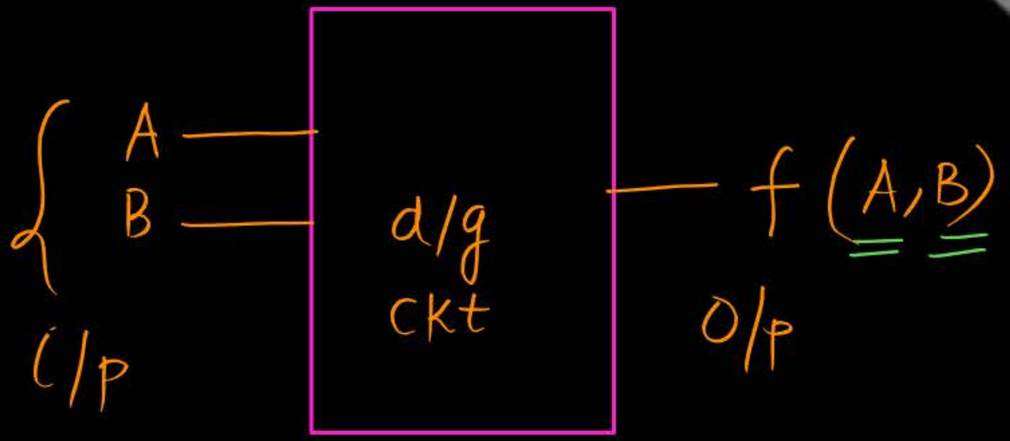
$= A\bar{B}+\bar{A}B$

$(A+B)(\bar{A}+\bar{B}) = A\bar{B}+\bar{A}B$

- $\overline{A}B + A\overline{C} = (\overline{A} + \overline{C})(A + B)$
- $A\overline{B} + B\overline{C} = (\overline{B} + \overline{C})(B + A)$

- $AB + B\overline{C} = B[A + \overline{C}]$

	2^1 A	2^0 B	SOP	POS	f
0	0	0	$\bar{A}\cdot\bar{B}$	$(A+B)$	1
1	0	1	$\bar{A}\cdot B$	$(A+\bar{B})$	0
2	1	0	$A\cdot\bar{B}$	$(\bar{A}+B)$	1
3	1	1	$A\cdot B$	$(\bar{A}+\bar{B})$	0



$$f(A,B) = \sum(0,2) = \bar{A}\cdot\bar{B} + A\cdot\bar{B} = \bar{B}\cdot 1 = \bar{B}$$

$$= \pi(1,3) = (A+\bar{B})\cdot(\bar{A}+\bar{B})$$

$$= \bar{B} + A\cdot\bar{A} = \bar{B}$$

$\bar{A}$ → 0
A → 1

A → 0
$\bar{A}$ → 1

SOP → sum of product form
term

POS → Product of sum → term form.

$$f(A,B,C) = \Sigma(1,2,4,6,7) = \pi(0,3,5)$$ ✓

3-variable $\rightarrow (0-7)$

$$f(A,B,C,D) = \Sigma(0,1,4,5,10,11,13)$$

$$= \pi(2,3,6,7,8,9,12,14,15)$$

A B C D

2^4 comb. $\rightarrow (0-15)$

$(1111)_2 \rightarrow (2^4-1)_{10}$

A B C D

term -10 $\longrightarrow$ SOP

$(1010)_2$ $\quad A\bar{B}\cdot C\cdot\bar{D}$

POS

$(\bar{A}+B+\bar{C}+D)$

A B C

term 7 $\longrightarrow$ SOP

$A\cdot B\cdot C$

POS

$(\bar{A}+\bar{B}+\bar{C})$

H.W

- $f(A,B,C,D) = \Pi(1,2,4,6,8,10,12) = \Sigma(?)$
- $f(A,B,C,D) = \Sigma(1,2,3,4,6,10,11) = \Pi(?)$

Write down the terms in SOP format

Write down the term in POS form.



2 Minute Summary

- Theorems & some basics.

Thank you

GW

Soldiers !

